

kubectl Quick Reference

Adapted from the official [Kubernetes kubectl Quick Reference](#), © The Kubernetes Authors, licensed under [CC BY 4.0](#). Reorganized here for the Practical Kubernetes course; command syntax preserved from source.

Autocomplete

```
source <(kubectl completion bash)          # bash
source <(kubectl completion zsh)           # zsh
echo 'kubectl completion fish | source' >>
~/.config/fish/completions/kubectl.fish # fish

kubectl get pods --all-namespaces # kubectl -A is shorthand for --all-
namespaces
```

Context and Configuration

```
kubectl config view                        # show merged
kubeconfig
kubectl config get-contexts                # list all
contexts
kubectl config current-context             # show current
context
kubectl config use-context <name>         # switch
context
kubectl config set-context --current --namespace=<ns> # set default
namespace
kubectl config set-credentials <user>    # add user
credentials
```

Apply (declarative — preferred for production)

```
kubectl apply -f ./manifest.yaml
kubectl apply -f ./dir # apply all
manifests in a directory
kubectl apply -f https://example.com/manifest.yaml
kubectl diff -f ./my-manifest.yaml # preview
changes before applying
```

Creating Objects (imperative)

```
kubectl create deployment nginx --image=nginx
kubectl create job hello --image=busybox:1.28 -- echo "Hello World"
kubectl create cronjob hello --image=busybox:1.28 --schedule="*/1 * * * *"
kubectl explain pods # field-level
docs for a resource
kubectl explain pods.spec.containers # drill into a
nested field
```

Viewing and Finding Resources

```
kubectl get services
kubectl get pods --all-namespaces
kubectl get pods -o wide
kubectl get pod <name> -o yaml
kubectl describe nodes <name>
```

Sorting & selecting:

```
kubectl get services --sort-by=.metadata.name
kubectl get pods --sort-by='.status.containerStatuses[0].restartCount'
kubectl get pods --selector=app=cassandra
kubectl get node --selector='!node-role.kubernetes.io/control-plane'
kubectl get pods --field-selector=status.phase=Running
```

JSONPath / custom output:

```
kubectl get pods --selector=app=cassandra -o
jsonpath='{.items[*].metadata.labels.version}'
kubectl get configmap myconfig -o jsonpath='{.data.ca.crt}'
kubectl get nodes -o custom-
```

```
columns='NODE_NAME:.metadata.name,STATUS:.status.conditions[?(@.type=="Ready")].status'
```

Events:

```
kubectl get events --sort-by=.metadata.creationTimestamp  
kubectl events --types=Warning
```

Updating Resources

```
kubectl set image deployment/nginx nginx=nginx:1.9.1  
kubectl rollout status deployment/nginx  
kubectl rollout undo deployment/nginx  
kubectl rollout undo deployment/nginx --to-revision=2  
kubectl rollout pause deployment/nginx  
kubectl rollout resume deployment/nginx
```

Patching & Editing

```
kubectl patch node k8s-node -p '{"spec":{"unschedulable":true}}'  
kubectl edit svc/docker-registry  
kubectl edit deployment/myapp
```

Scaling

```
kubectl scale --replicas=3 rs/foo  
kubectl scale --replicas=3 -f foo.yaml  
kubectl scale --current-replicas=2 --replicas=3 deployment/mysql  
kubectl autoscale deployment nginx --min=10 --max=15
```

Deleting Resources

```
kubectl delete pod,service baz foo  
kubectl delete pods,services -l name=myLabel  
kubectl delete pod foo --now
```

```
kubectl delete pod foo --grace-period=0 --force
kubectl delete deployment/foo
```

Interacting with Running Pods

```
kubectl logs my-pod
kubectl logs my-pod --previous          # previous
container instance
kubectl logs my-pod -c my-container
kubectl logs -f deployment/my-app --all-containers=true  # stream all
containers
kubectl run -i --tty busybox --image=busybox -- sh
kubectl attach my-pod -i
kubectl port-forward pod/my-pod 5000:6000
kubectl exec my-pod -- ls /
kubectl exec --stdin --tty my-pod -- /bin/bash
```

Copying Files

```
kubectl cp /tmp/foo_dir my-pod:/tmp/bar_dir
kubectl cp my-pod:/tmp/foo /tmp/bar
kubectl cp my-namespace/my-pod:/tmp/foo /tmp/bar
```

Deployments & Services

```
kubectl set image deployment/frontend www=image:v2
kubectl rollout status -w deployment/frontend
kubectl rollout undo deployment/frontend
kubectl get replicaset
kubectl get pods --selector=app=nginx
kubectl annotate pods my-pod icon-url=http://example.com/icon.png
kubectl autoscale deployment foo --min=2 --max=10
```

Nodes & Cluster

```
kubectl cordon my-node          # prevent
scheduling
kubectl drain my-node          # gracefully
```

```
terminate pods
kubectl uncordon my-node # resume
scheduling
kubectl top nodes
kubectl top pods --all-namespaces
kubectl cluster-info
kubectl cluster-info dump
```

Common Resource Types (with short names)

Resource	Short name
pods	po
services	svc
deployments	deploy
replicasets	rs
statefulsets	sts
daemonsets	ds
jobs	—
cronjobs	—
configmaps	cm
secrets	—
namespaces	ns
nodes	no
persistentvolumes	pv
persistentvolumeclaims	pvc
ingresses	ing

Output Formatting

```
-o json # JSON
-o yaml # YAML
```

```
-o wide                # additional columns
-o name                # resource name only
-o custom-columns=<spec> # custom columns
-o jsonpath=<template> # JSONPath template
```

Verbosity / Debugging

```
--v=0  # always visible
--v=1  # reasonable default
--v=2  # useful steady-state info
--v=3  # extended info about changes
--v=4  # debug level
--v=5  # trace level
--v=6 --alsologtostderr # log to file and stderr
```